Choose your operating system

One setup, one package environment and one shared image for JFEM and its apps.

Download and extract the complete repository, then open `__Start here!__`. Choose the folder matching your computer. Each folder has one guide, one setup launcher and four application launchers.

Folder	Setup	Instructions
Windows	setup.cmd	Windows/README.md
Linux	setup.sh	Linux/README.md
Mac	setup.command	Mac/README.md

The existing launchers in the repository root remain available. Keep the whole checkout together in a writable location.

In this guide: Windows is illustrated on the following pages; Linux and macOS have dedicated command pages. Repair and offline preparation are on the final page.

Ready when: you have chosen the folder for your operating system.

What opens after installation

Application	Purpose
JFEM-Boxer	Lifting-surface models, loads, FE analysis, sensitivity and results.
PANDEATOR	Stiffened-panel modeling and analysis.
Results-viewer	Open existing .jfem result files.
JFEM-solver	Choose a deck and run the solver directly.

Setup installs the packages and builds the shared local image automatically. The first build can take an hour or longer, including long quiet periods during native code generation. Keep the setup console open until it finishes.

Julia, the packages and the image are shared by the solver, apps and workers. New model specializations may still compile on first use.

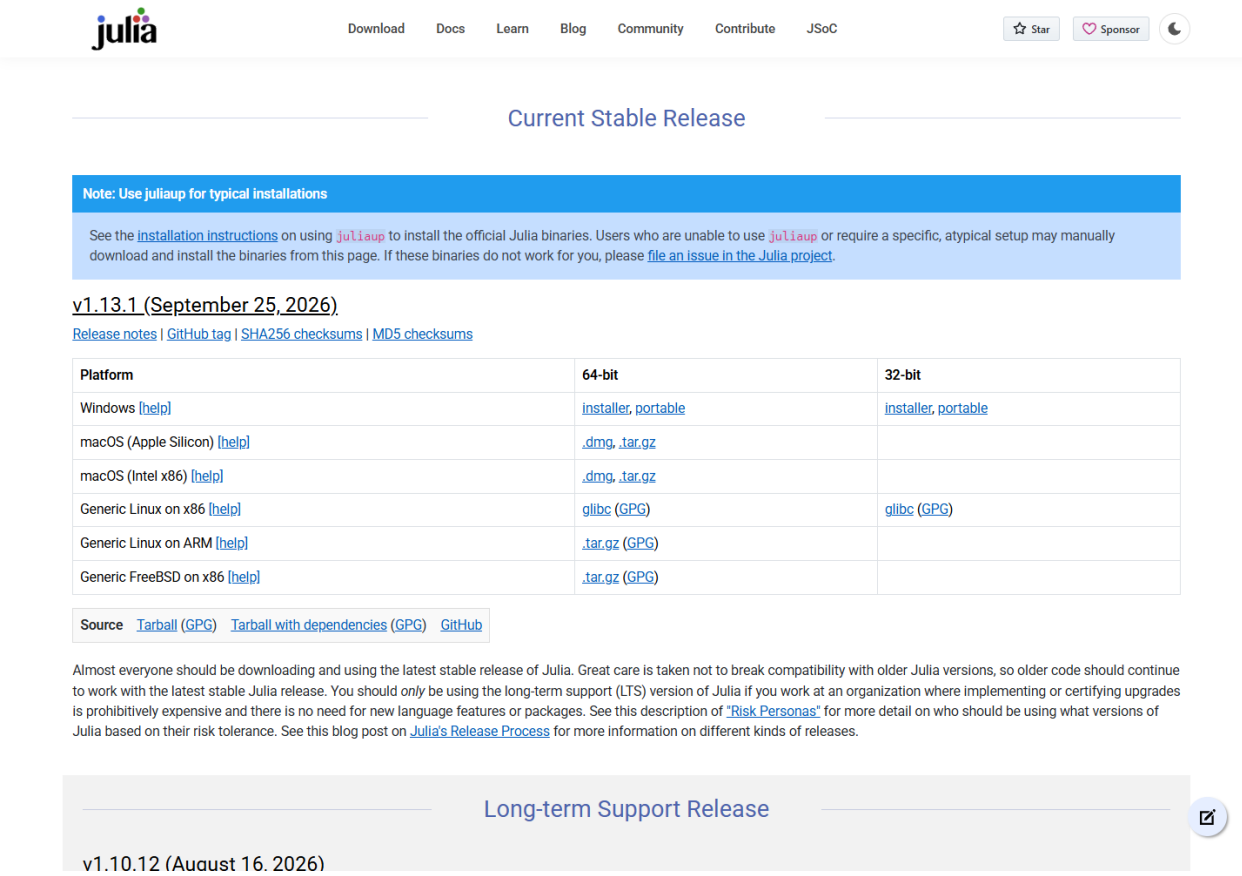
1 Windows: install Julia

Tick Add Julia to PATH. The launchers need it to find Julia.

1. Open julialang.org/downloads/manual-downloads.
2. Download the Windows 64-bit installer for **Julia 1.13.1**, the release tested with this checkout.
3. Run the installer and **tick Add Julia to PATH**. Do not skip this option.
4. Finish installation. Reopen any windows that were already open before installing Julia.

If Julia is already installed without PATH, rerun/modify its installer or follow the official Windows installation instructions.

Ready when: the Julia installation is complete and its executable is on PATH.



Choose the installer for your operating system. The webpage appearance may change.

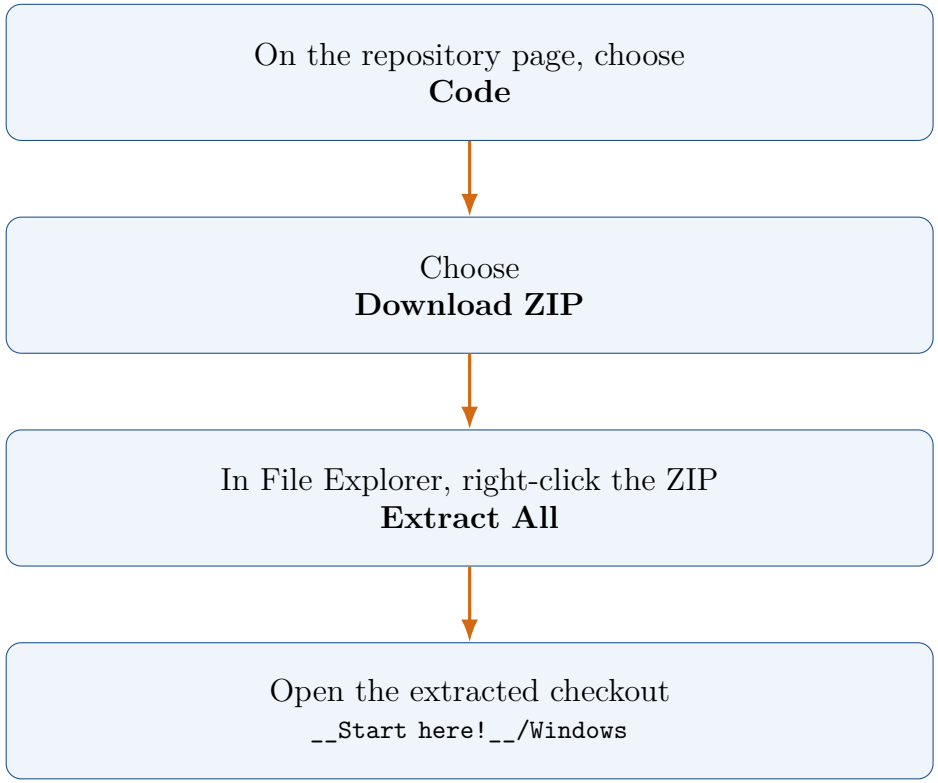
2 Download and extract JFEM

Keep the entire repository together. Do not run it inside the ZIP.

1. Open github.com/r-acad/JF_2026_05_25.
2. Click **Code**, then **Download ZIP**.
3. Right-click the download and choose **Extract All**.
4. Open the extracted folder, then open `__Start here!__/Windows`.

Choose a writable location. App folders use the shared solver, packages and assets beside them.

Ready when: you are looking at ordinary extracted files, not the ZIP contents.



Download and extract the complete repository before running setup.

3 Double-click setup.cmd

One setup installs the packages and builds one shared sysimage automatically.

1. Double-click **setup.cmd** in `__Start here!__/Windows`. No Julia command is needed.

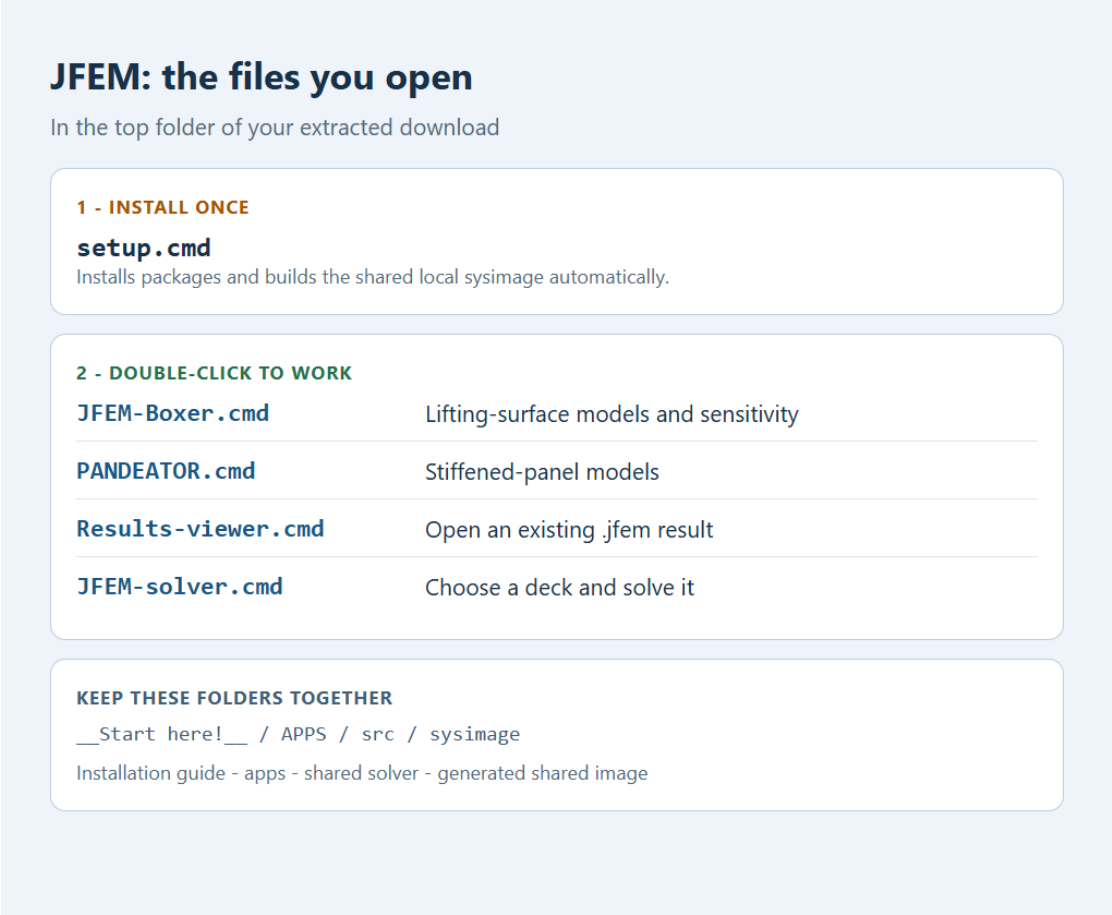
2. Stay connected to the internet. Setup installs packages, checks the apps and precompiles representative workloads.

3. Wait while it builds the **shared JFEM/app sysimage**. The solver and all apps use the same root Project/Manifest and package versions. The first build can take an hour or longer, with long quiet periods during code generation.

4. Keep the window open until setup reports completion. If an error is reported, retain its text and rerun setup after addressing it.

Repeated setup reuses a valid image. No manual package list is required.

Ready when: setup has finished successfully, including the shared local image.



The original root launchers shown here remain valid shortcuts. The same launchers are now grouped in the Windows folder.

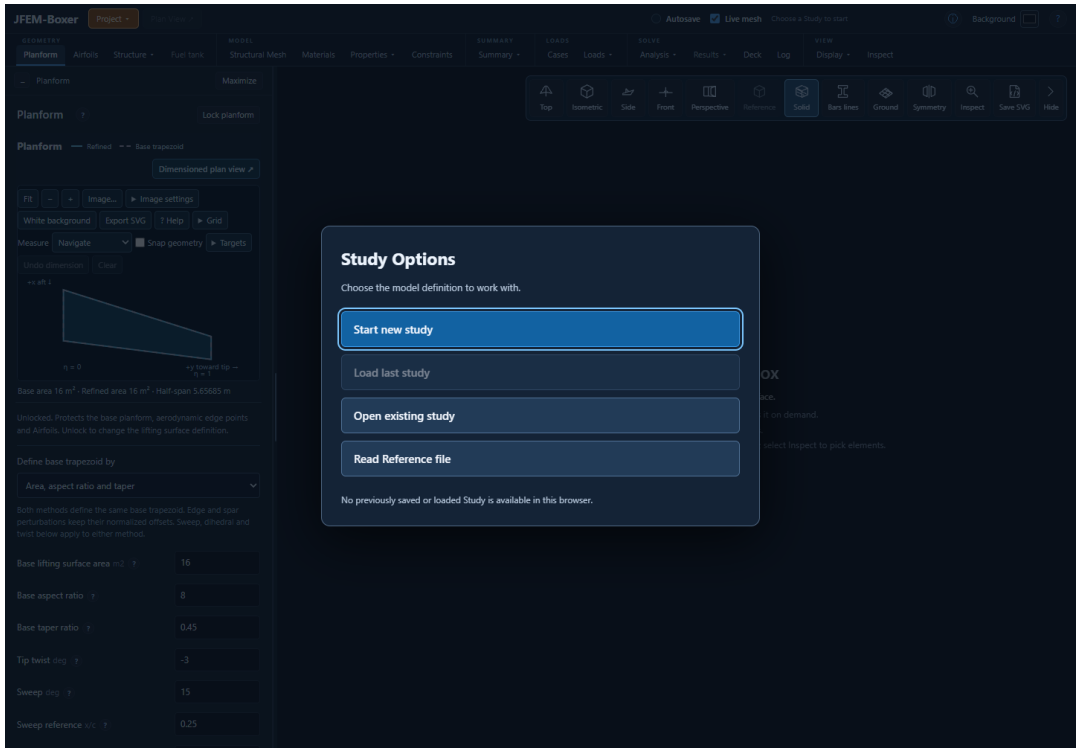
4 Double-click your application

Open a launcher in the Windows folder. Root shortcuts remain available.

File	Purpose
JFEM-Boxer.cmd	Lifting-surface modeling, loads, FE analysis, sensitivity and results.
PANDEATOR.cmd	Stiffened-panel modeling and analysis.
Results-viewer.cmd	Open existing .jfem results; no Julia server.
JFEM-solver.cmd	Choose an input deck and run JFEM directly; drag-and-drop also works.

In JFEM-Boxer choose **Start new study**, or use **Project** to reopen a saved study or import a Reference deck. Keep the JFEM-Boxer/PANDEATOR console open while working: it runs the local server. The browser address is printed there.

Ready when: the chosen app is open.



JFEM-Boxer starts with Study Options. Choose the appropriate model source.

5 Run and inspect a model

The case, field, deformation scale and legend explain what is being displayed.

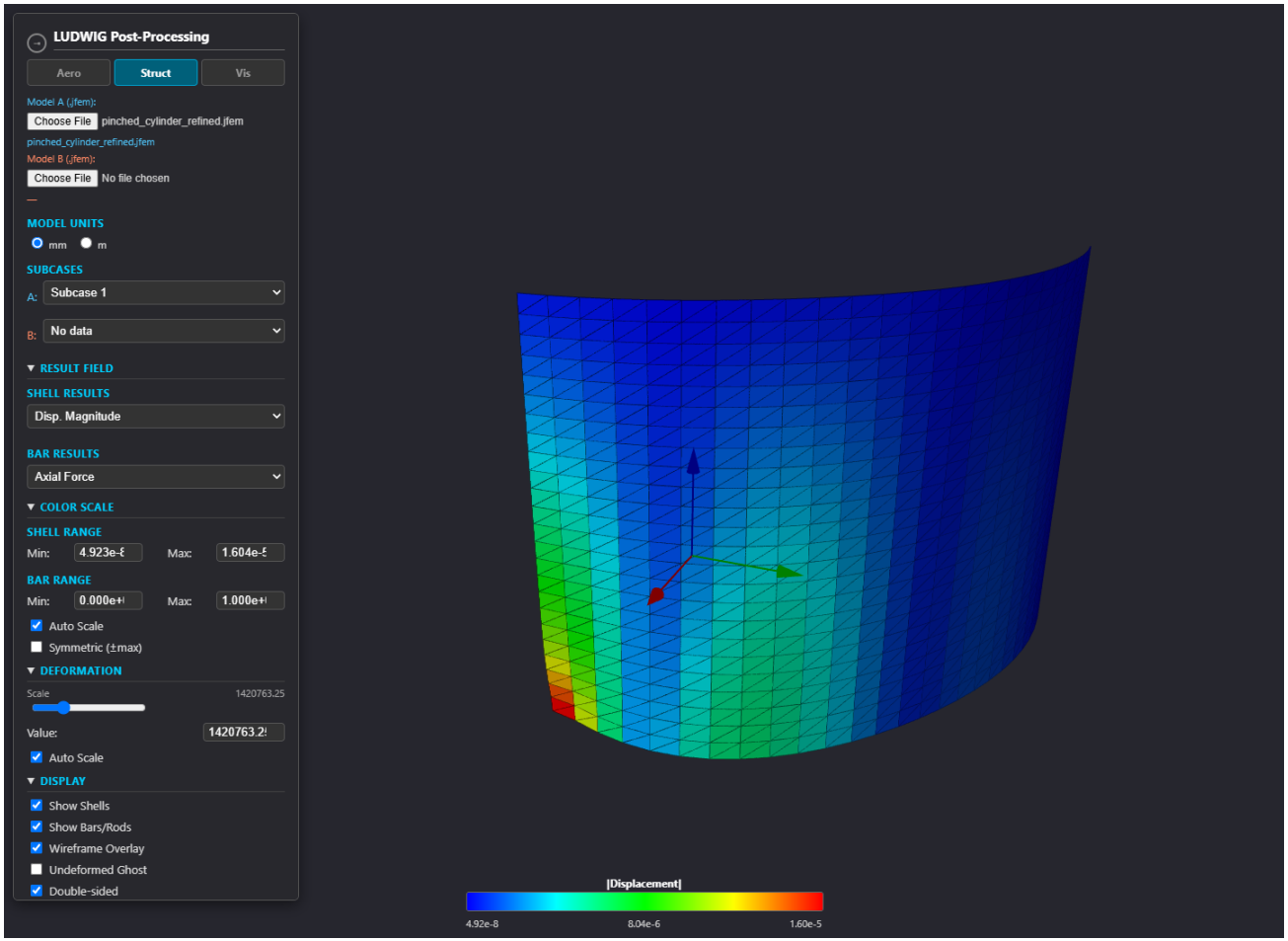
In JFEM-Boxer create the mesh, then use **Analysis / FE Analysis**. Define a solution and click **Run in JFEM**. Sensitivities are under **Analysis / Sensitivity**.

To try standalone JFEM, double-click **JFEM-solver.cmd** and select:

`utils/examples/precompile/sol101_quad_static.bdf`

The console reports the output location. Open **Results-viewer.cmd**, select **Struct**, then **Choose File** for Model A and load the resulting `.jfem`.

Choose the case, result field and deformation scale. The viewer reads results; it does not solve models.



Worked example: the public refined pinched-cylinder model (576 shells), displacement magnitude, mesh edges and an automatically amplified deformation. The field legend shows the actual values; amplification is only visual.

Linux

Open a terminal in `__Start here!__/Linux`.

1. Install **Julia 1.13.1** for your processor architecture from the official Julia downloads page.

2. Make `julia` available on PATH. Open a new terminal and check `julia --version`.

3. Download and extract the complete repository to a writable directory.

4. Open a terminal in `__Start here!__/Linux` and run setup:

```
bash setup.sh
```

Stay connected until setup succeeds. It builds the shared image automatically; the first build can take an hour or longer. Keep the terminal open during quiet code-generation periods.

Ready when: setup reports successful completion.

Launch an application from the same folder:

```
bash JFEM-Boxer.sh
bash PANDEATOR.sh
bash Results-viewer.sh
bash JFEM-solver.sh
```

Keep the Boxer/PANDEATOR terminal open while using its browser page. Press **Ctrl+C** there when finished. If the browser does not open, use the local address printed in the console.

The solver uses a graphical deck chooser when available, otherwise a terminal prompt. You can also pass a deck directly:

```
bash JFEM-solver.sh '/path/to/model.bdf'
```

The viewer opens existing results using bundled browser assets. Its HTML file can also be opened directly from `APPS/JFEM_results_viewer/postv11.html`.

macOS

Use `__Start here!__/Mac` for Finder launchers and Terminal commands.

1. Install **Julia 1.13.1** for your Mac's processor architecture. Make `julia` available on PATH.
2. Confirm `julia --version` in a new Terminal window.
3. Download and extract the complete repository to a writable directory.
4. Open `__Start here!__/Mac` and double-click **setup.command**.

If executable permissions were not retained by the ZIP, open Terminal in that Mac folder and run:

```
chmod +x ./*.command
bash setup.command
```

Stay online until setup succeeds. The shared image is built automatically; the first build can take an hour or longer, including quiet code-generation periods.

Double-click one of these files in the Mac folder:

- **JFEM-Boxer.command**
- **PANDEATOR.command**
- **Results-viewer.command**
- **JFEM-solver.command**

Terminal works too:

```
bash JFEM-Boxer.command
bash JFEM-solver.command '/path/to/model.bdf'
```

The solver opens a deck chooser when no path is given. Keep the Boxer/PANDEATOR Terminal open while using the local browser app; press **Ctrl+C** to stop it. Root application shortcuts remain available and use the same environment.

Ready when: your chosen app is open and its console remains available.

Repair, updates and offline use

The same root environment and verified shared image serve every app.

1. **JFEM-Boxer, standalone JFEM, PANDEATOR and analysis/sensitivity workers** load `sysimage/OpenJFEM_sysimage.dll`.

2. The root Project/Manifest pins the same packages for every app. No separate application environment or solver checkout is needed.

3. Launchers check source, dependencies and Julia version before loading an image. The console reports the selected image.

4. After updating the checkout, Julia or packages, or moving the folder, rerun the setup launcher for your OS. Do not force an old image with `-J`.

The image avoids repeating compilation covered by the installation workloads. New models or code paths can still require first-use compilation. Linux images use `.so`; macOS uses `.dylib`. The image and its receipt stay local and are excluded from Git. Each installation builds its own cache for its Julia version, platform and packages. Shared examples, images, Julia helpers, the Python client and tests live in `utils/`, beside `src/`.

Repair Restore network/proxy access and rerun setup after a download failure. If an image build runs out of memory, close other applications and use fewer build threads. Do not delete your studies or input decks.

```
.\setup.cmd --dry-run
.\setup.cmd --sysimage-threads=1
.\setup.cmd --no-sysimage
```

Use `bash setup.sh` on Linux or `bash setup.command` on Mac with the same switches. The last option explicitly skips image construction and retains normal source loading.

Offline preparation Complete setup on a connected computer with the same OS, architecture and Julia release. Copy the full repository and Julia depot (normally `%USERPROFILE%/.julia` on Windows or `~/.julia` on Linux/Mac). Install matching Julia on the target, put it on PATH, and run its setup launcher with `--offline`. Missing packages/artifacts require refreshing the cache on the connected machine. Keep the Manifest unchanged.

Copyable OS instructions: `__Start here!__/README.md`. Solver commands and automation: `utils/julia_tools/README.md`. The JFEM-Boxer guide is under `Reference_documentation/00_Getting_Started/`.